

A Search For Rapid Rotation In Massive White Dwarf Stars

Fall 2024 Joint Meeting of TSAPS, TSAAPT, and Zone
13 SPS at Southern Methodist University

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University: TAMUC



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COMMERCE



Outline

- Background
- Goal
- Target Selection
- Observations
- Methods
- Analysis
- Results



Image Credit: TAMUC

Stellar Evolution

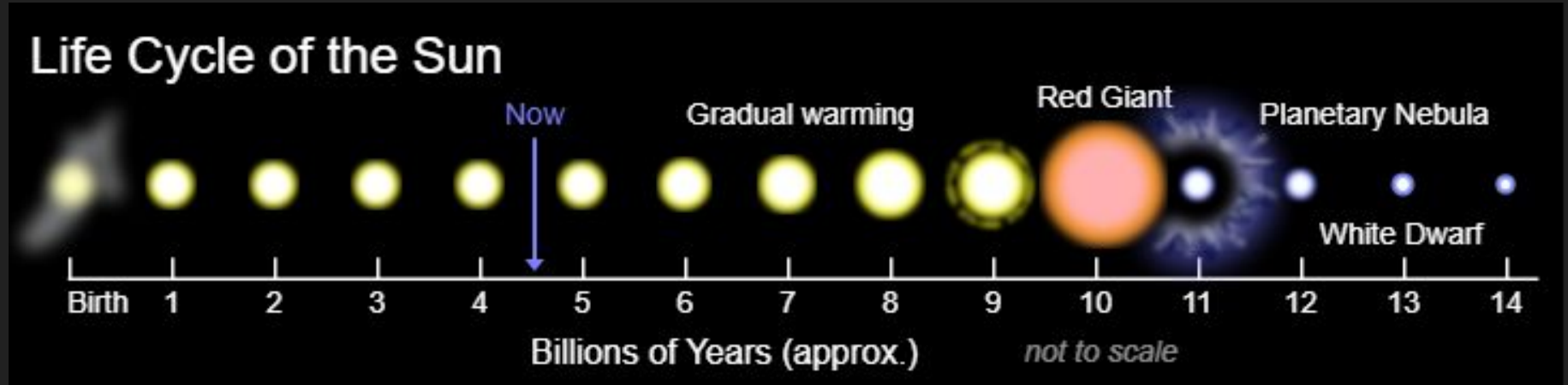


Image Credit: David Jeffery; www.physics.unlv.edu

White Dwarf Stars (WDs)

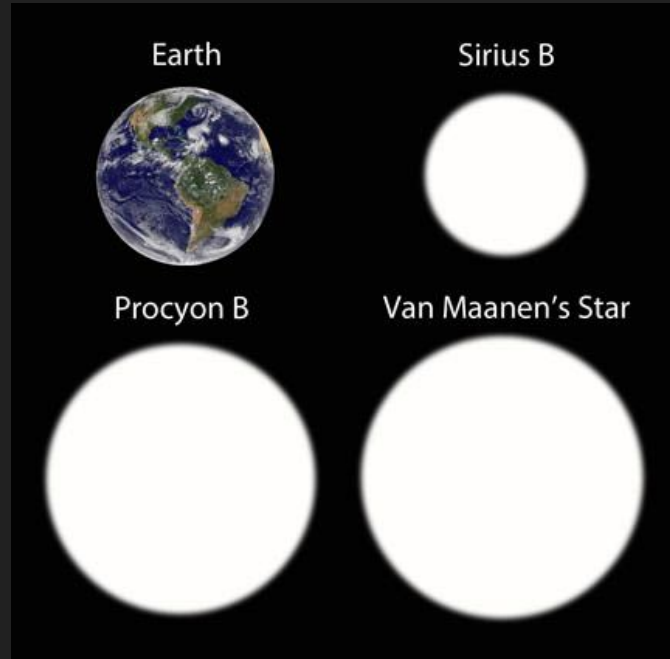


Image Credit: Bob King; skyandtelescope.org

My Project

Goal: Search for variation and rapid rotation in massive WDs.

- Why might it vary in luminosity?
 - Starspots
- Why might it rapidly rotate?
 - Massive Sun-like star
 - Double Degenerate Merger
- What is rapid rotation?
 - Period ≤ 6 hours

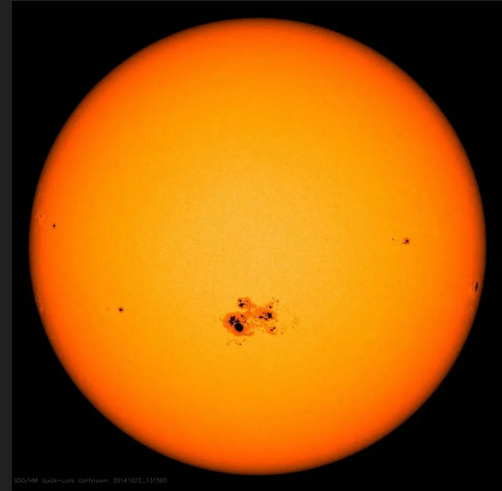


Image credit: NASA/SDO

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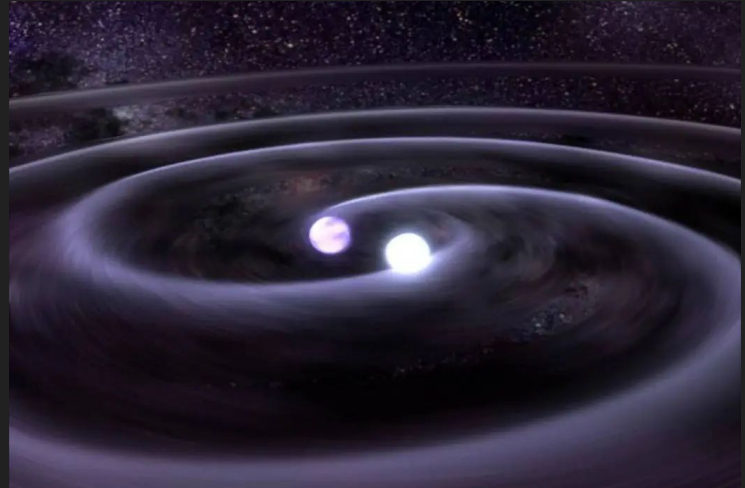
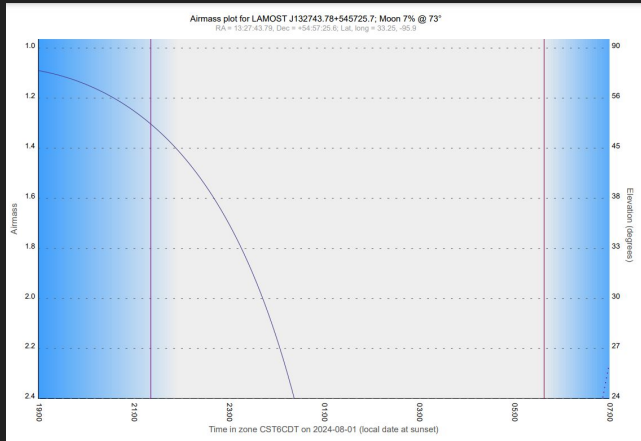


Image Credit: NASA/Tod Strohmayer (GSFC)/Dana Berry (Chandra X-Ray Observatory)

Target Selection

- Montreal White Dwarf Database
- MAST TESS Archive
- Airmass Plot



Montreal White Dwarf Database

Home Tables and charts References Database description Evolutionary tracks

Search filters

	Minimum	Maximum	Spectral Type
T_{eff} [K]:		T_{eff} [K]:	☒ DA 12
$\log(g)$:		$\log(g)$:	☒ DB 0
distance [pc]:		distance [pc]:	☒ DC 0
Mass v	0.9	G v 16.5	☒ DQ 1
			☒ DZ 0
			☒ DO 0
			☐ PG1159 0
			☒ CND ² 0
			☐ Subdwarf 0
			☐ Other 0

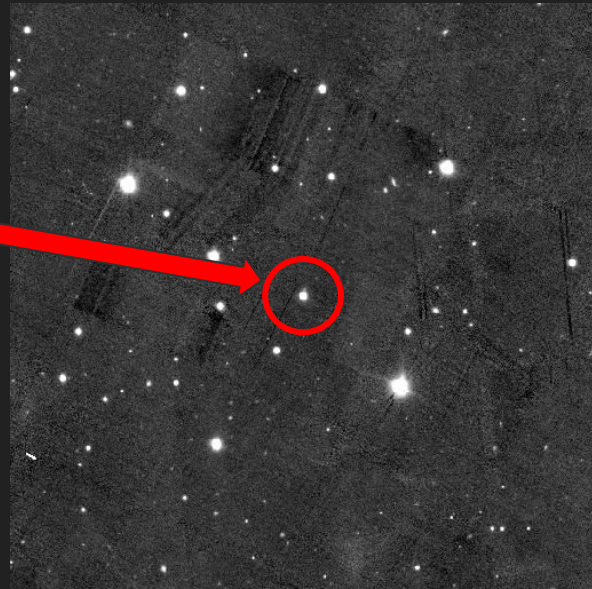
Total: 13
(13 including unresolved double degenerates)

Radial search

Right ascension	Declination	Distance
hh mm ss.ss	dd mm ss.ss	arcmin

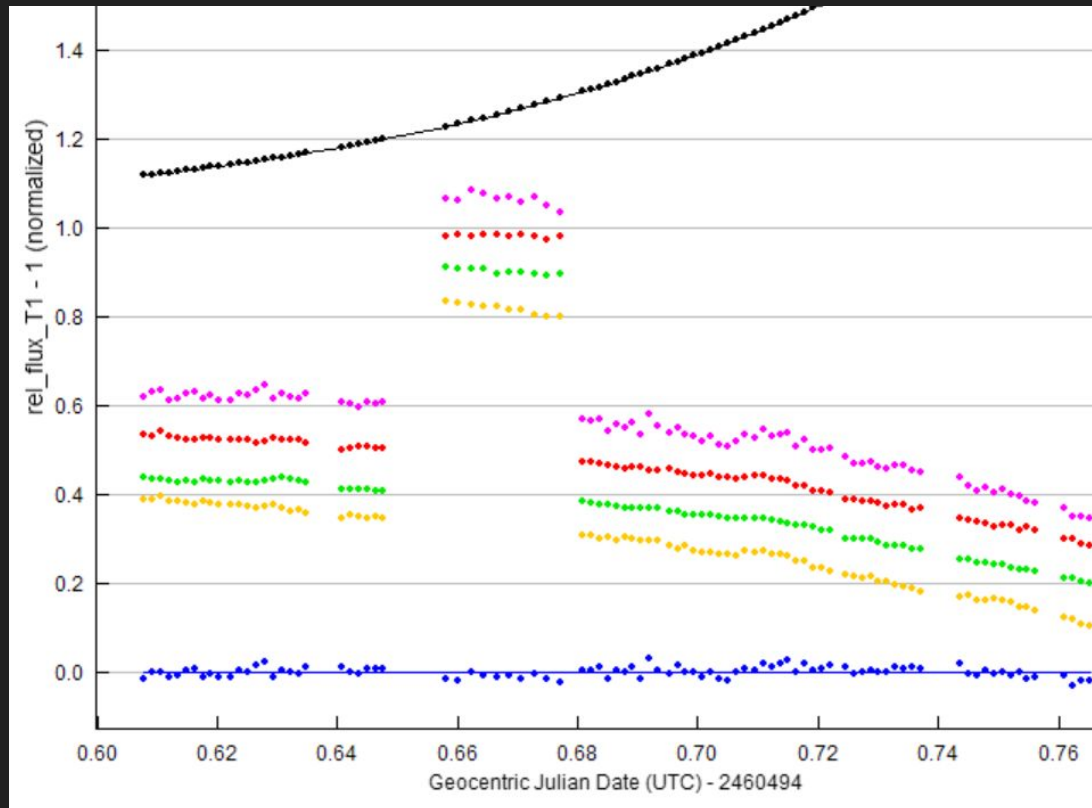
Stars Observed

- PG 1533-057
 - Observed 2 days
 - $M \approx 0.94 M_{\odot}$
- LAMOST J132743.78+545725.7
 - Observed 4 days
 - $M \approx 1.05 M_{\odot}$
- WD 1459+347
 - Observed 2 days
 - $M \approx 0.925 M_{\odot}$



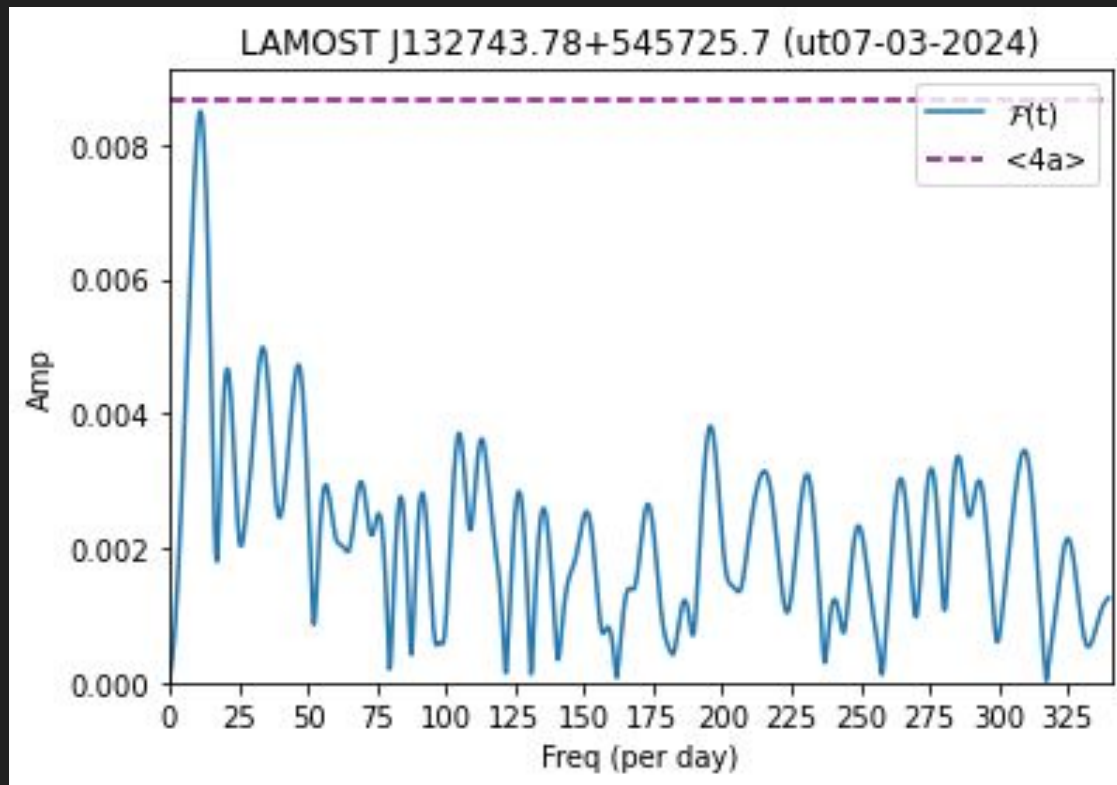
Info: MWDD, Image Credit: PanSTARRS-1

Photometry - Universal Time (UT) July 3rd

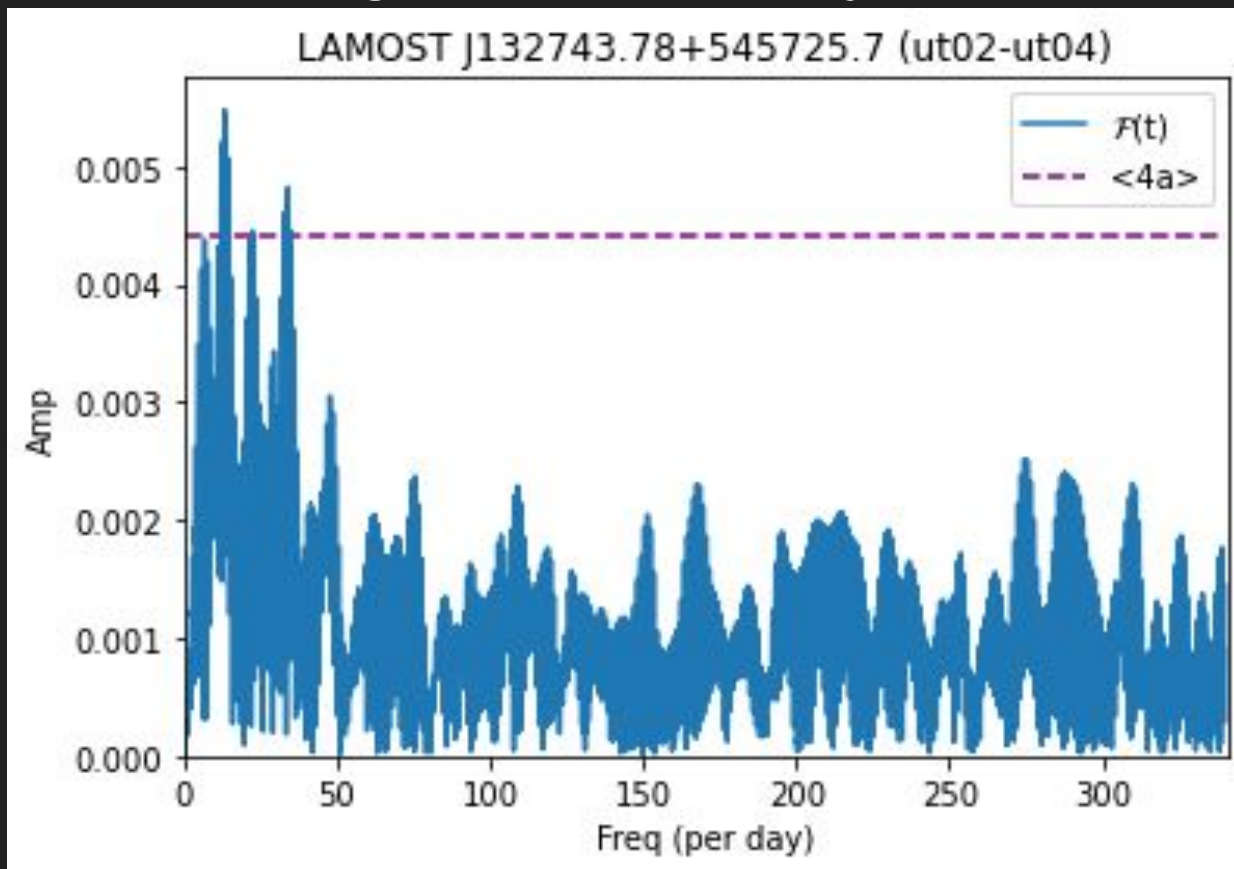


Periodogram - UT July 3rd

- Freq: 11.5 /day
- Period: 2.09 hours
- Amp: 0.008

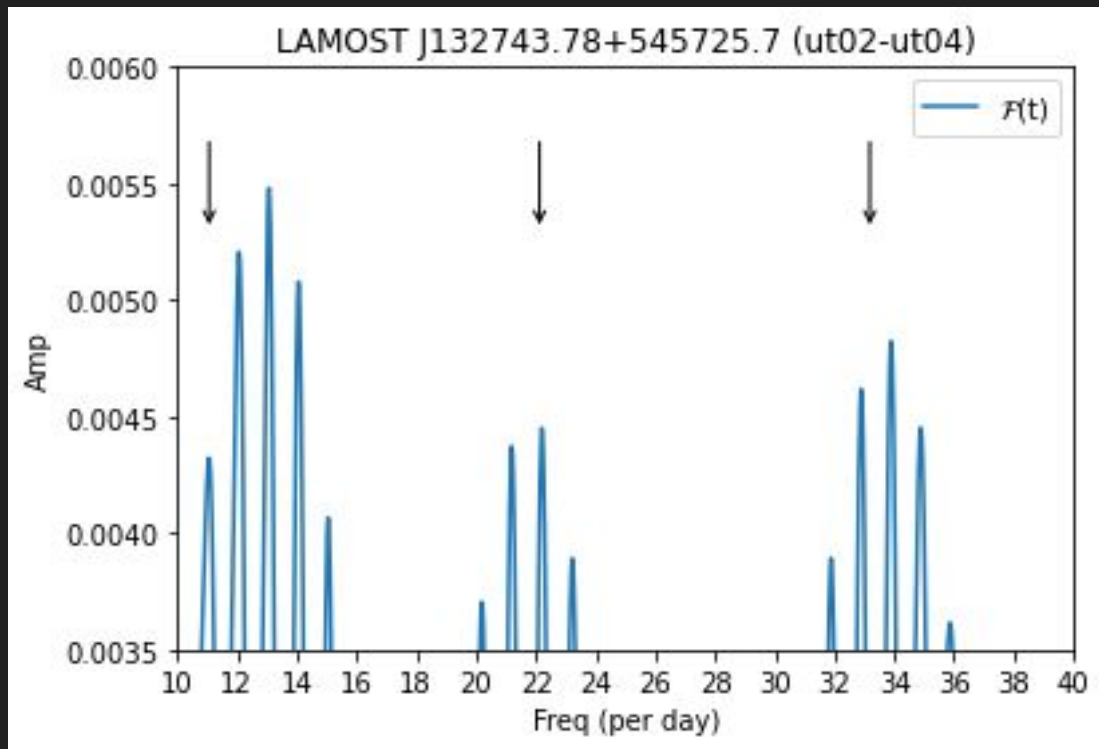


Periodogram - UT July 2nd-4th

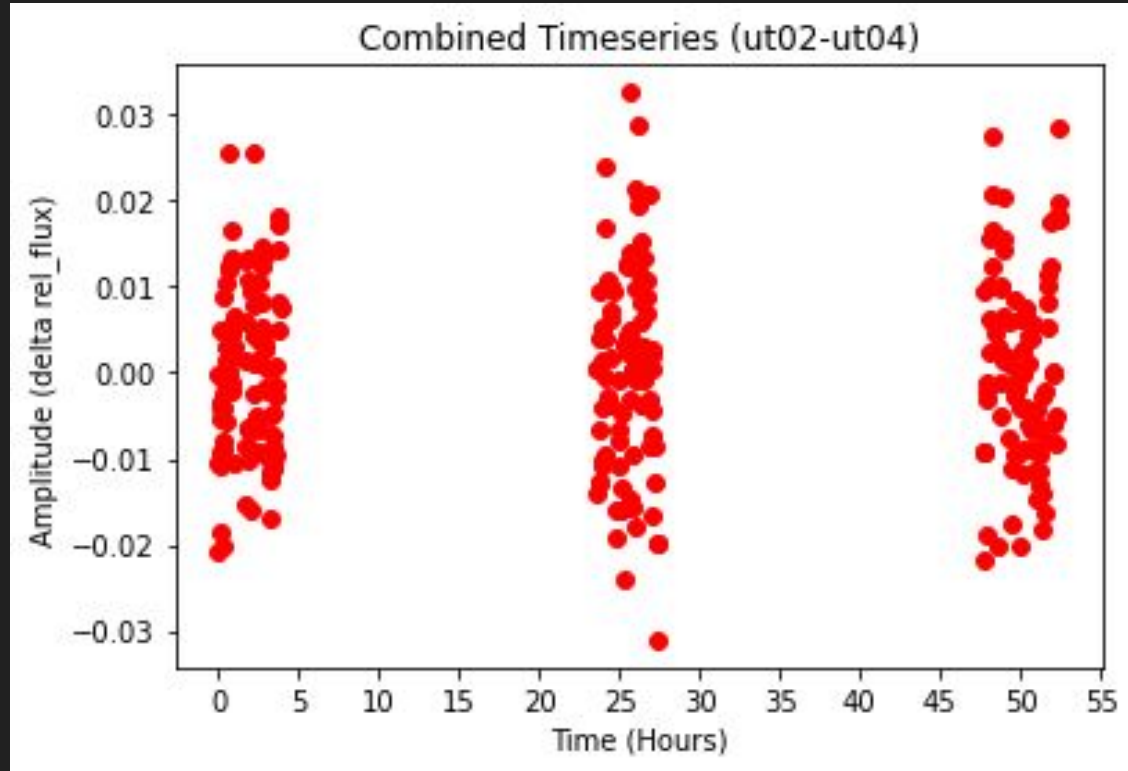


Periodogram - UT July 2nd-4th

- Alias 1
 - Freq: 11.06 /day
 - Period: 2.17 hours
 - Amp: 0.004
- Alias 2
 - Freq: 12.06 /day
 - Period: 1.99 hours
 - Amp: 0.005
- Harmonics
 - Second ≈ 22 /day
 - Third ≈ 33 /day

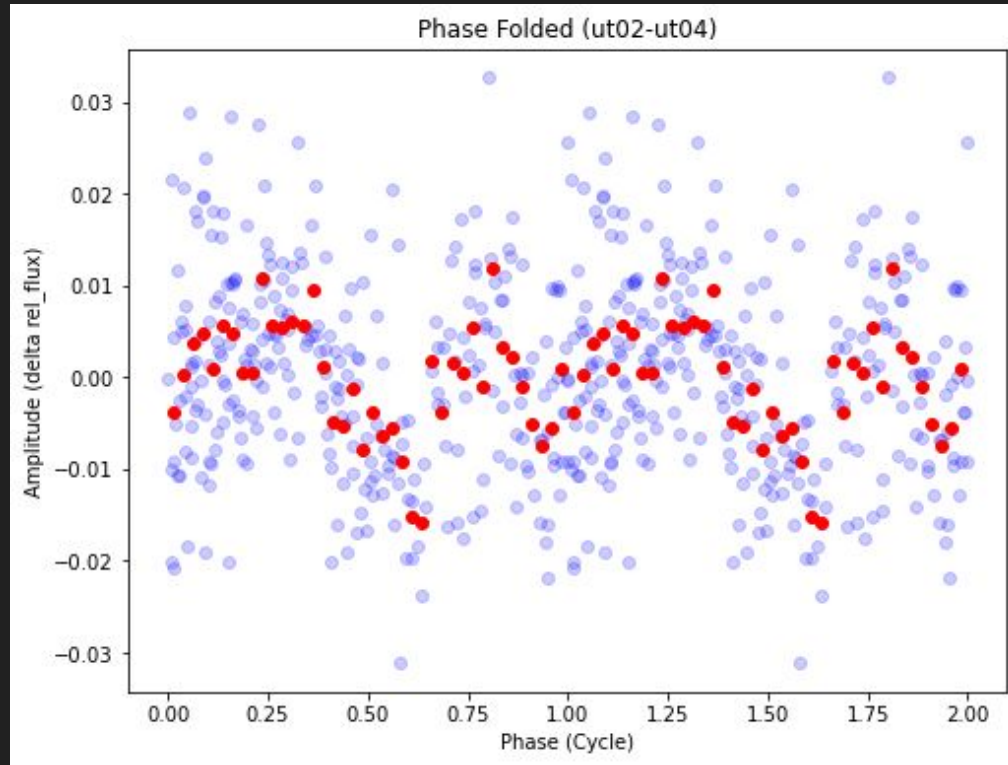


Combined Data - UT July 2nd-4th



Source Code Used: Dr. Matt Wood

Phase Fold Diagram - UT July 2nd-4th



Source Code Used: Dr. Matt Wood

Results

- PG 1533-057
 - Wasn't able to reduce data UT 06-12-2024
- LAMOST J132743.78+545725.7
 - Frequency: 11.06 ± 0.03 /day
 - Period: 2.170 ± 0.006 hours
 - Amplitude: 0.0043
- WD 1459+347
 - Not observed to vary at greater than 2% (Amp less than $<4\sigma$)

Acknowledgments



Dr. Williams



My Peers



Joshua Belieu



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Department of
Physics and Astronomy

